

Forecasting Bitcoin Realized Volatility

Econometric models versus machine learning in a rolling out-of-sample study

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Abstract

We forecast next-day realized variance of BTC/USDT, computed from 5-minute Binance returns over 2019–2026. Naive, GARCH(1,1) and HAR benchmarks are compared with XGBoost and an LSTM on 1,338 out-of-sample days, using a rolling 1,000-day window, the QLIKE loss and Diebold–Mariano tests. HAR models built on intraday data dominate GARCH, and a weekend dummy lowers QLIKE by a further 15%. Machine-learning models reduce QLIKE by about 16% relative to HAR+weekend, but a linear HAR fed with the same inputs is statistically indistinguishable from XGBoost ($p = 0.73$): the gain comes from feature engineering, chiefly intraweek seasonality, rather than from non-linearity.

1 Data and realized variance

We use Binance spot klines for BTC/USDT at a 5-minute frequency, from January 2019 to August 2026. Let $P_{t,i}$ be the close of the i -th 5-minute bar of day t (UTC) and $r_{t,i} = \log P_{t,i} - \log P_{t,i-1}$, with $M = 288$ bars per day. The daily return and the daily realized variance are

$$r_t = \sum_{i=1}^M r_{t,i}, \quad \text{RV}_t = \sum_{i=1}^M r_{t,i}^2. \quad (1)$$

Under a continuous semimartingale model $d \log P_s = \mu_s ds + \sigma_s dW_s$, RV_t converges to the integrated variance $\int_{t-1}^t \sigma_s^2 ds$ as $M \rightarrow \infty$ (Andersen et al., 2003), so it is a far more precise proxy of daily variance than r_t^2 . Days with fewer than 90% of the bars are dropped, which leaves about 2,790 days.

Stylized facts. (i) RV_t is extremely skewed (skewness ≈ 21 , excess kurtosis ≈ 615), whereas $y_t = \log \text{RV}_t$ is close to Gaussian (skewness 0.13, excess kurtosis 0.85); all models therefore target y_{t+1} . (ii) Returns are nearly unpredictable (first-order autocorrelation -0.07). (iii) y_t has long memory: its autocorrelation is 0.70 at lag 1 and still 0.27 at lag 30. (iv) The autocorrelation of y_t peaks every 7 days: although Bitcoin trades continuously, volatility is markedly lower on weekends, when traditional markets and institutional desks are closed.

2 Models

Naive. $\widehat{\text{RV}}_{t+1} = \text{RV}_t$.

GARCH(1,1). On daily returns (Bollerslev, 1986),

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad (2)$$

with z_t standardized Student- t , estimated by maximum likelihood; the forecast is $\sigma_{t+1|t}^2$.

HAR family. The heterogeneous autoregressive model (Corsi, 2009) approximates long memory with averages over three horizons. With $y_t^{(w)} = \frac{1}{7} \sum_{j=0}^6 y_{t-j}$ (7 days, since crypto trades every day) and $y_t^{(m)} = \frac{1}{30} \sum_{j=0}^{29} y_{t-j}$,

$$y_{t+1} = \beta_0 + \beta_d y_t + \beta_w y_t^{(w)} + \beta_m y_t^{(m)} + \gamma^\top D_{t+1} + \varepsilon_{t+1}, \quad (3)$$

estimated by OLS. *HAR* has no calendar term; *HAR+weekend* uses a single dummy D_{t+1} , equal to one when day $t+1$ is a Saturday or a Sunday; *HAR+dow* uses six weekday dummies (Monday as reference); *HAR-X* additionally includes $y_{t-1}, \dots, y_{t-6}$, the vol-of-vol (7-day standard deviation of y), r_t , $|r_t|$ and $\min(r_t, 0)$ (leverage effect), i.e. exactly the inputs given to the ML models.

Machine learning. *XGBoost* (Chen & Guestrin, 2016): gradient-boosted trees of depth 3, learning rate 0.03, subsampling 0.8, up to 1,000 trees with early stopping. *LSTM* (Hochreiter & Schmidhuber, 1997): one layer of 32 units on standardized 30-day sequences of the same features, Adam with weight decay 10^{-4} , early stopping (patience 20). *Ensemble*: equal-weight average of the *HAR+weekend*, *XGBoost* and *LSTM* variance forecasts.

From log to variance. Every model except Naive and GARCH predicts $\hat{\mu}_{t+1} = \hat{\mathbb{E}}[y_{t+1} | \mathcal{F}_t]$. If $y_{t+1} | \mathcal{F}_t \sim \mathcal{N}(\mu, s^2)$ then $\mathbb{E}[\text{RV}_{t+1} | \mathcal{F}_t] = e^{\mu + s^2/2}$, so we report $\hat{\text{RV}}_{t+1} = \exp(\hat{\mu}_{t+1} + \hat{s}^2/2)$, with $\hat{s}^2$ the residual variance (in-window for linear models, on the validation block for ML models). Omitting this correction biases forecasts downwards, which QLIKE penalizes heavily.

3 Forecasting protocol

The out-of-sample period runs from 1 January 2023 to 30 August 2026 (1,338 days). All models use a rolling window of the last 1,000 days, so that every forecast of RV_{t+1} uses only information available at the end of day t . *HAR* models are refit every day, *GARCH* every 20 days and ML models every 60 days; for the latter, the last 200 days of each window serve for early stopping and for $\hat{s}^2$.

Loss functions. Besides MSE and MSE on logs, the main criterion is

$$\text{QLIKE}(\text{RV}, h) = \frac{\text{RV}}{h} - \log \frac{\text{RV}}{h} - 1 \geq 0, \quad (4)$$

which is zero iff $h = \text{RV}$. Up to constants it is the negative Gaussian log-likelihood of a return with variance h . It depends only on the ratio RV/h , so crisis days do not dominate the average as they do under MSE, and it penalizes under-prediction more than over-prediction (a forecast of half the true variance costs 0.31, twice the true variance 0.19). Crucially, QLIKE and MSE are the only standard losses that rank forecasts consistently when the target is a noisy but unbiased proxy of the true variance (Patton, 2011).

Diebold–Mariano test. For two models with losses $L_t^{(1)}, L_t^{(2)}$, let $d_t = L_t^{(1)} - L_t^{(2)}$ and $\bar{d}$ its mean over n days. Then

$$DM = \frac{\bar{d}}{\sqrt{\hat{\Omega}/n}}, \quad \hat{\Omega} = \hat{\gamma}_0 + 2 \sum_{h=1}^5 \left(1 - \frac{h}{6}\right) \hat{\gamma}_h, \quad (5)$$

where $\hat{\gamma}_h$ are sample autocovariances of d_t (Newey–West). Under the null of equal predictive accuracy, DM is asymptotically $\mathcal{N}(0, 1)$ (Diebold & Mariano, 1995); with rolling-window estimation this remains valid for comparing nested models (Giacomini & White, 2006). A negative statistic means that model 1 is more accurate.

4 Results

	Estimate	Std. error	t
Constant	−0.511	0.131	−3.9
y_t (daily)	0.340	0.019	18.2
$y_t^{(w)}$ (weekly)	0.479	0.032	15.1
$y_t^{(m)}$ (monthly)	0.088	0.030	2.9
Weekend dummy	−0.619	0.029	−21.4

Table 1: HAR+weekend estimated on the full sample (interpretation only), $R^2 = 0.62$.

Table 1 confirms the stylized facts. The persistence $\beta_d + \beta_w + \beta_m = 0.91$ is high, with the weekly component dominating. The weekend coefficient implies $e^{-0.62} \approx 0.54$: variance is about 46% lower on weekends, i.e. volatility about 27% lower, and it is the most significant regressor.

Model	QLIKE	vs Naive	MSE log	DM vs HAR+weekend	DM vs XGBoost
Naive	0.5433	—	0.7916	+8.71 (<0.001)	—
GARCH(1,1)	0.4159	−23%	1.0444	+4.16 (<0.001)	—
HAR	0.3466	−36%	0.6597	+5.58 (<0.001)	—
HAR+weekend	0.2943	−46%	0.4751	—	+1.40 (0.161)
HAR+dow	0.2688	−51%	0.3969	−3.83 (<0.001)	+0.57 (0.567)
HAR-X	0.2586	−52%	0.3777	−4.67 (<0.001)	+0.35 (0.728)
XGBoost	0.2477	−54%	0.4578	−1.40 (0.161)	—
LSTM	0.2606	−52%	0.4286	−1.16 (0.245)	+0.70 (0.483)
Ensemble	0.2485	−54%	0.4342	−1.90 (0.057)	+0.07 (0.941)

Table 2: Out-of-sample results, 2023-01-01 to 2026-08-30 (1,338 days). DM: Diebold–Mariano statistic on QLIKE, p -value in parentheses.

Three conclusions emerge from Table 2.

- 1. Intraday information beats daily returns.** Every HAR variant dominates GARCH. GARCH only sees daily returns, a very noisy signal, so its forecasts are smooth and miss both spikes and the weekly cycle; its MSE on logs is even worse than the naive forecast.
- 2. Calendar effects are the largest single gain.** The weekend dummy cuts QLIKE by 15% relative to HAR, and weekday dummies by a further 9%, both highly significant.
- 3. The ML advantage is a feature effect, not a non-linearity effect.** XGBoost reaches the lowest QLIKE (−16% vs HAR+weekend) and the ensemble the lowest MSE, but neither is significantly better than HAR+weekend at the 5% level. XGBoost’s most important features by gain are the weekend indicator, the next day of the week, the weekly average and y_{t-6} (same weekday one week earlier), i.e. calendar information. Once the linear model receives the same inputs (HAR-X), it significantly beats HAR+weekend, obtains the best MSE on logs, and is statistically

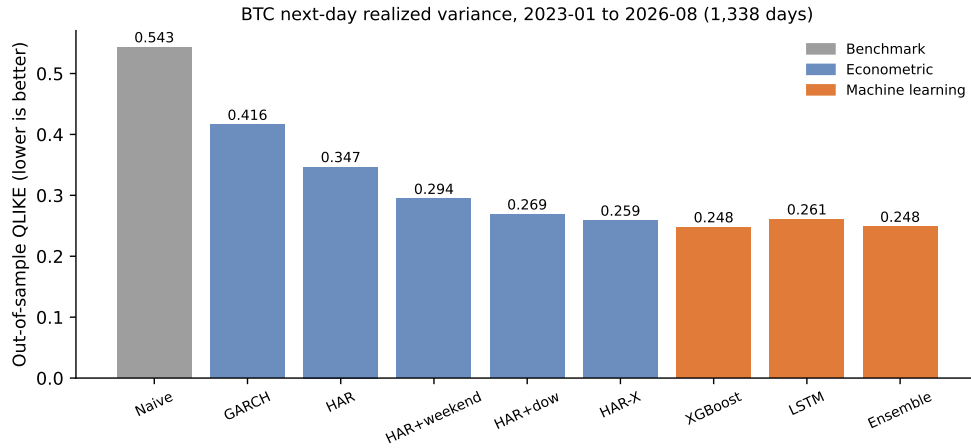


Figure 1: Out-of-sample QLIKE by model.

indistinguishable from XGBoost ($p = 0.73$), while remaining interpretable and essentially free to estimate.

5 Limitations and extensions

The study covers a single asset and a one-day horizon, with fixed ML hyperparameters. Five-minute sampling limits but does not remove microstructure noise; realized kernels or subsampling would be more robust, and jump-robust measures (bipower variation) would allow HAR-J or HARQ specifications (Bollerslev et al., 2016). Pairwise DM tests could be replaced by a Model Confidence Set (Hansen et al., 2011). Natural extensions are multi-horizon forecasts (1, 5 and 22 days), other crypto-assets, and an economic evaluation, for instance volatility targeting or Value-at-Risk backtests driven by each forecast.

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